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Rugby Union

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Performance Demands of Rugby Union

In this section, we introduce the sport of Rugby Union and discuss the various factors of importance for players (and individual positions) and the relative contributions that physical performance characteristics make to winning games.

Sport Description and Factors of Winning

Rugby Union, commonly known as rugby, is a contact team sport that originated in England in the first half of the 19th century. Rugby Union is an invasion sport that requires mostly running with ball in hand with the game objective of scoring more points than the opposition. Points can be scored in several ways. A try, scored by grounding the ball over the try-line in the in-goal area of the opposition (between the try-line and the dead-ball line), is worth 5 points and a subsequent conversion kick scores 2 points. A successful penalty kick or a drop goal each scores 3 points. Kicks are required to pass between two goalposts and over a connecting cross-bar. Rugby is a game of territory and possession with players allowed to pass or kick the ball. The most common form of rugby is a game between two teams

of 15 players using an oval-shaped ball on a rectangular field (100 m long and 70 m wide) with H-shaped goalposts situated centrally on each try-line. Rugby is a global sport with about 9 million players registered in 2016 and over 100 ranked and accredited nations. Rugby boasts the second most lucrative and most attended sporting World Cup, with the 2015 Rugby World Cup selling 2.47 million tickets and generating ~\$500 million in revenue.

The player positions are shown in figure 28.1. Positions typically are classified into two main groups, the forwards and backs. Forwards consist of front rowers (props and hooker; numbers 1 to 3), second row (locks; numbers 4 and 5), and the back row (or loose forwards; numbers 6 to 8). The backs consist of inside backs (halfback, also called a Scrum-half, first 5/8, second 5/8; numbers 9 to 12) and outside backs (center, wings, and fullback; numbers 13 to 15).

Rugby Union games are divided into 40 min halves, separated by a 10 to 15 min half-time break. Stoppages for injury or to allow the referee to take disciplinary action do not count as part of the playing time, so that the elapsed time is usually longer than 80 min (often approximately 100 min). Rugby Union is an intermittent sport characterized by periods of intense activity such as sprints, tackles, accelerations, decelerations, rucks, mauls, and scrums, interspersed with low-intensity periods

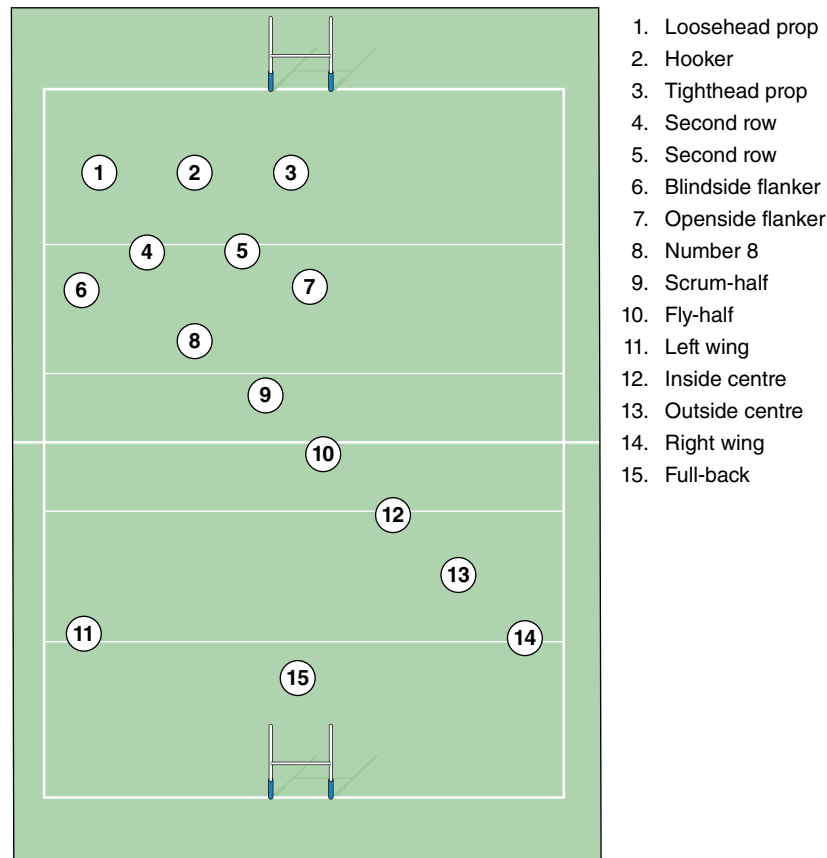


Figure 28.1 The positions of Rugby Union.

primarily consisting of walking and jogging (5, 7, 28). Research using time-motion analysis and wearable tracking technology such as a global positioning system (GPS) shows activity profiles that vary between playing positions (15, 21). Like many sports, intensity is typically shown to increase as the competition standard rises (5, 21). As well, during match play, backs tend to cover a greater total distance, more high-intensity running distance (6 to 8 m/s), and very high-speed running distance (>8 m/s) than forwards (1, 21, 25). The positional requirements of forward play require that these players perform higher volumes of high-intensity activity (22), repeated high-intensity efforts (15), and contact loads (21) than backs.

Passing and Kicking

The ball in rugby can be passed laterally or backward but not forward. The ball tends to be moved forward in three ways including by kicking, by having a player run with it, or by moving the ball within a scrum or maul (see below). Only players in possession of the ball may be tackled.

Any player may kick the ball forward in an attempt to gain territory. When a player anywhere in the playing area kicks the ball into touch (across the sideline), a lineout is taken. Here, the opposite team that kicked the ball into touch gets to throw the ball into a lineout (see below).

Breakdowns

Tackles in rugby must be below the neck, with the aim of impeding or grounding the player with the ball. The aim of the defending side is to stop the player with the ball, either by bringing him to the ground (termed a *ruck*) or by contesting for possession with the ballcarrier on his feet (termed a *maul*). Such a circumstance is called a breakdown, and specific laws govern each situation.

Tackling

A player can tackle an opposing player who has the ball by holding him while bringing him to the ground. However, it is illegal to push, shoulder-charge, or trip a player using the feet or legs, although hands can be used.

Lineout

As mentioned before, when the ball leaves the side of the field (either through a kick, pass, or a player being tackled out across the sideline), a lineout is awarded against the team that last touched the ball. Forward players from each team line up a meter apart, perpendicular to the touchline and between 5 m and 15 m from the touchline. The team's hooker will use an overhead throw to toss the ball down the middle of the lines created by the two teams. One or more players from each team will be lifted by their teammates in order to try to gain possession of the ball from the air (24). Both sides compete for the ball and players have the option of lifting their teammates. A jumping player cannot be tackled until he lands, and only shoulder-to-shoulder contact is allowed.

Scrum

A scrum is a way of restarting the game safely and fairly after a minor infringement. Scrums are awarded when the ball has been knocked or passed forward, if a player takes the ball over his own try-line and puts the ball down, when a player is accidentally offside, or when the ball is trapped in a ruck or maul with no realistic chance of being retrieved. A team may also opt for a scrum if awarded a penalty.

A scrum is formed by the eight forwards from each team (see figure 28.1) binding together in three rows. The front row consists of the two props and a hooker. The second row consists of two locks and the back row consists of two flankers and a number 8. Once a scrum is formed, the scrum-half from the team awarded the scrum rolls the ball into the gap between the two opposing front rows. The two hookers then compete for possession by hooking the ball backward with their feet, while each pack tries to push the opposing pack backward to help gain possession.

Relative Contribution of Physical Performance

Possessing certain physical attributes is important for Rugby Union players, but is of limited value if the athlete does not possess the specific positional skills and game understanding, which are fundamental to performance. In conjunction with time-motion analysis, a parallel stream of research within Rugby Union has been the use of notational analysis to quantify the physical and skill requirements of

competition. Notational analysis provides objective feedback of games and player actions through the frequencies of key performance indicators (9, 16). Specifically, notational analysis has been used to determine differences in playing patterns between teams and individuals and assist in understanding what leads to successful performance (10, 14, 16). It is clear from existing data that as the level of competition gets higher, the positional skills and understanding of the game improves. In saying this, however, so too do the physical capabilities of the athletes.

When it comes to the match, the team that is likely to win tends to be the team that can execute the game plan at the highest intensity level for the game duration. This means the positional skills need to be accurate, the understanding of the game plan needs to be sound, and players must have the physical capacity to execute all required tasks at a high intensity for 80 minutes. At the very least, rugby players need to be fit enough to tolerate the requirements of their positions and have the ability to execute their roles under fatigue, under pressure, and within many dynamic and fast-changing situations. Notwithstanding the varying requirements of the multiple Rugby Union positions shown (figure 28.1; described below), a rugby player generally requires very high levels of strength and the ability to repeatedly and effectively express that strength and power over the entire game duration (figure 28.2). Players maintain heart rates close to 90% of maximum (4), and hence, HIIT becomes a vital component in the arsenal of the conditioning program designed to enhance metabolic aspects and repeated high-intensity requirements vital to Rugby Union success.

Targets of Physical Performance in Rugby Union

The physical characteristics of Rugby Union players and the differences between positions and playing levels are well documented. Essentially, forwards are typically heavier and stronger than backs, while backs are faster and more agile than forwards (19). The differences between positions are related to the differences in the position-specific tasks performed during a match. For example, time-motion analysis and the use of GPS have shown backs travel farther, sprint farther and more frequently, and have lower work:rest ratios than forwards (1, 4, 6, 7).

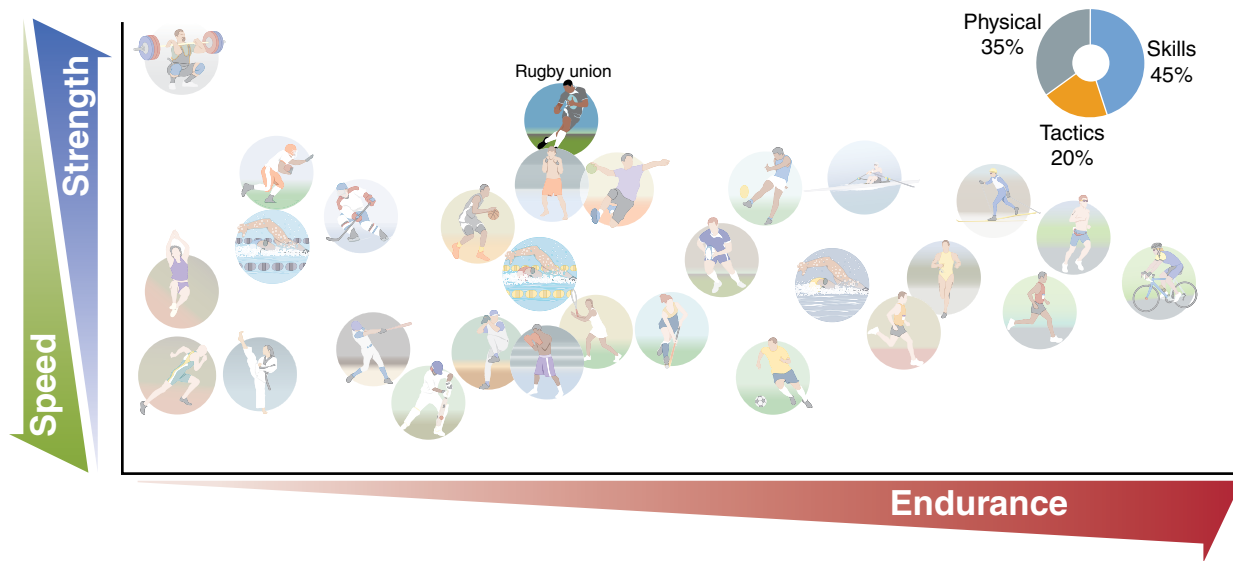


Figure 28.2 The position of the Rugby Union player on the three axes illustrates the relative importance of the three main physical capacities of importance for elite participation, acknowledging that this will vary between positions. Outside of critical psychological aspects and team chemistry components, which are difficult to quantify, the pie chart shows the general relative importance of skills (45%), tactical awareness (20%), and physical capacities (35%) for Rugby Union success.

Adapted from G.A. Nader, "Concurrent Strength and Endurance Training: From Molecules to Man," *Medicine & Science in Sports & Exercise* 38, no. 11 (2006): 1965-1970.

Tight Forwards

The front row and second row play vital roles in securing possession of the ball in the restarts, line-outs, and scrum situations during the game. Due to the required tasks of these players, strength and power are very important assets for pushing, jumping, and tackling. While these players are also typically the heaviest members of the team, the need for aerobic fitness is also important, as they will typically cover 4 to 6 km in a game. Offensively and defensively these players are in the closer channels to the high-contact ruck and maul situations and can make between 10 and 30 tackles a game. Lean muscle mass, strength, and power are therefore crucial physical attributes for these players to perform their core roles.

Back Row

Back row players are typically involved in running with the ball and tackling in slightly wider channels than tight forwards. While these players still have an important role in the restarts, lineouts, and scrums, they also have a role in linking the tight forwards with the backs and getting into a number of positions quickly to have influence on the game. Turning over the ball (stealing from the opposition)

and supporting line breaks are an important role and requires speed, power, and a high level of anaerobic and aerobic metabolic conditioning.

Inside Backs

Numbers 9, 10, and 12 (figure 28.1) are typically smaller than most other players and typically are the kickers of the team. Their tactical nous and game understanding is typically very high as they are called on to drive the team around the field. Due to the natural link between the forwards and the outside backs, these players must normally possess a balance of speed, power, endurance, and agility.

Outside Backs

As the outside backs have a major role in finishing plays, counterattacking, and chasing kicks, these players are typically explosive and very fast runners of the ball. Typically the leanest players on the field, they often cover the most distance in a game (especially the fullback). Often distances of sprints for these players are longer than all other players on the field.

Match Demands

Austin et al. (1) found that the work:rest ratios of Super 14 Rugby Union players were 1:4, 1:5, and 1:6

for front row and back row forwards, inside backs, and outside backs, respectively. During game play, players cover substantial distance and are involved in multiple high-intensity running and nonrunning activities (1, 11, 13, 20). The demands of professional Rugby Union matches between the years 2008 and 2015 were recently reported by Hogarth and colleagues (13), who showed that the distance covered by players during a match ranged from 4662 to 7227 m, depending on playing position and the level of competition. The 33 to 39 min ball-in-play duration of a match and the total distance covered by players suggest that there is a significant demand on a player's aerobic system to provide the required energy to perform the necessary activities over this period of time (8). During most competition matches, the duration of each play (whistle to whistle) typically depends on the level of competition, but lasts for 35 s on average, although some periods of play can last up to 6 min.

In addition to requiring a well-conditioned aerobic system, a significant demand is placed on the player's anaerobic system due to the intermittent short high-intensity running and nonrunning activities that occur throughout a match (4, 6, 8). Austin et al. (1) found that the passive:high-intensity task work ratio (nonrunning:scrum, ruck, maul) varied the most between the forwards and backs, with front rowers and back rowers performing the most of these tasks (62 ± 13 and 68 ± 15 , respectively) and the inside and outside backs completing only one-third of this type of work (17 ± 7 and 14 ± 5), respectively.

Key Weapons, Manipulations, and Surveillance Tools

Developing and monitoring both endurance and high-intensity running abilities are important considerations for many team sport athletes. The enhancement of these qualities can positively impact performance during the repeated bouts of high-intensity activity that occur in professional Rugby Union competition (2, 17, 27). For example, a moderate correlation ($r = -0.38$) has been shown between match activity rate (tasks/min) and repeated sprint ability (23), while increased aerobic capacity, as determined by maximal aerobic running speed, is correlated ($r = 0.746$) to increased match running distance in professional Rugby Union players (25). This evidence supports strategies aimed at enhance-

ment of aerobic endurance and high-intensity running capacity in professional rugby players. However, it is important to acknowledge the complex interplay of physical, technical, tactical, environmental, and cognitive factors that influence competitive match performance (figure 28.2), which makes it difficult to directly relate match performance to specific outcomes on physical tests.

HIIT Weapons

We typically target a range and combination of HIIT weapons throughout the year depending on the individual needs, training phase (preseason, in-season, etc.), turnaround time between games, technical and tactical training load, and intensity, training surface, weather, and travel factors. Game-based high-intensity interval training (GBHIIT), generally broken down into small-sided games (SSGs) and skill-based games (SBGs), as well as short and long intervals (on-feet and off-feet) and mixed-fuel circuits, form the basis of conditioning content for Rugby Union players. The general percentage breakdown of our HIIT weapons over the course of a season is shown in figure 28.3.

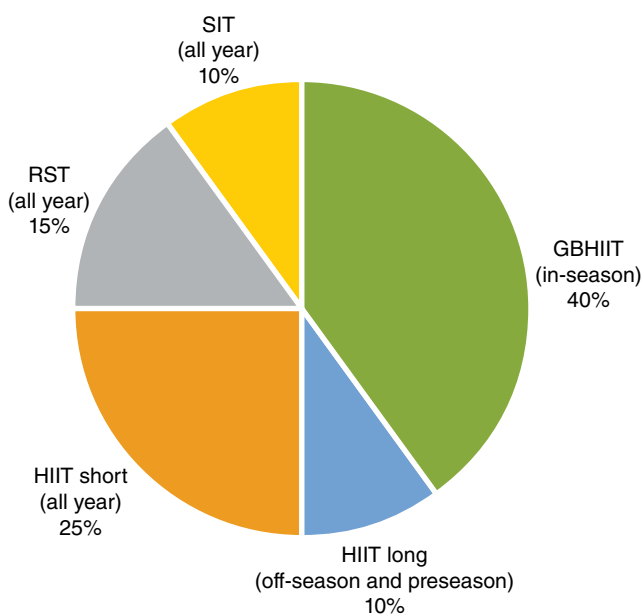


Figure 28.3 Percentage of the different HIIT formats (weapons) used throughout the annual season in elite Rugby Union preparation. Note that a 12 mo cycle consists of approximately 30 games, and the exact breakdown depends on the level of competition encountered, the duration of competition, and the positional and individual needs.

Manipulations of Interval Training Variables

The running intensity, inclusion of static work, duration of effort, and work:rest ratios are the main variables manipulated to achieve the desired stress for the particular purpose of any given HIIT session in Rugby Union player preparation. Positional needs, individual needs, and specific metabolic and neuromuscular focus are considered for each session, each week, and during each macrocycle.

GBHIIT

Due to the overall importance of skill and game understanding in Rugby Union, it is vital to incorporate these game-based factors into the conditioning program. For many teams, GBHIIT is the main HIIT weapon of choice! Essentially, this incorporation is due to a number of reasons, but the fact that skill- or game-based conditioning can be an efficient way to improve game-specific skill and understanding for groups or teams, while also requiring periods of high-intensity work, makes it a staple weapon for most teams (12), although it is acknowledged that GBHIIT may not always adequately mimic the demands of competitive matches (26). Rugby Union players, like most athletes, are highly competitive,

so utilizing this competitive drive helps to ensure that athletes work hard when there is a ball involved. Get them individually or in a line to do some tough intervals, and the motivation for many athletes is not quite the same. In addition, the benefit of having a bit of fun while working hard together cannot be underestimated in terms of interpersonal relationships and team building.

SSGs are an effective training modality and are increasingly used across team sports in an attempt to elicit a physiological response, while simultaneously challenging technical, tactical, and decision-making aspects of performance. For example, analysis of various soccer-specific SSGs showed these to be reliable and effective training measures across sessions in terms of running output, and they can be manipulated through alterations in various constraints such as field size and number of players (3, 18). With regard to alterations in GBHIIT, volume, and intensities, it typically comes down to the purpose of the session and training or competition phase, but adjustments in field size, player numbers, rules, and so forth enable us to deliver the running load, metabolic load, and skill acquisition aspects required to challenge and progressively load athletes. To be clear, we don't necessarily want to be hitting match intensity all the time (see also

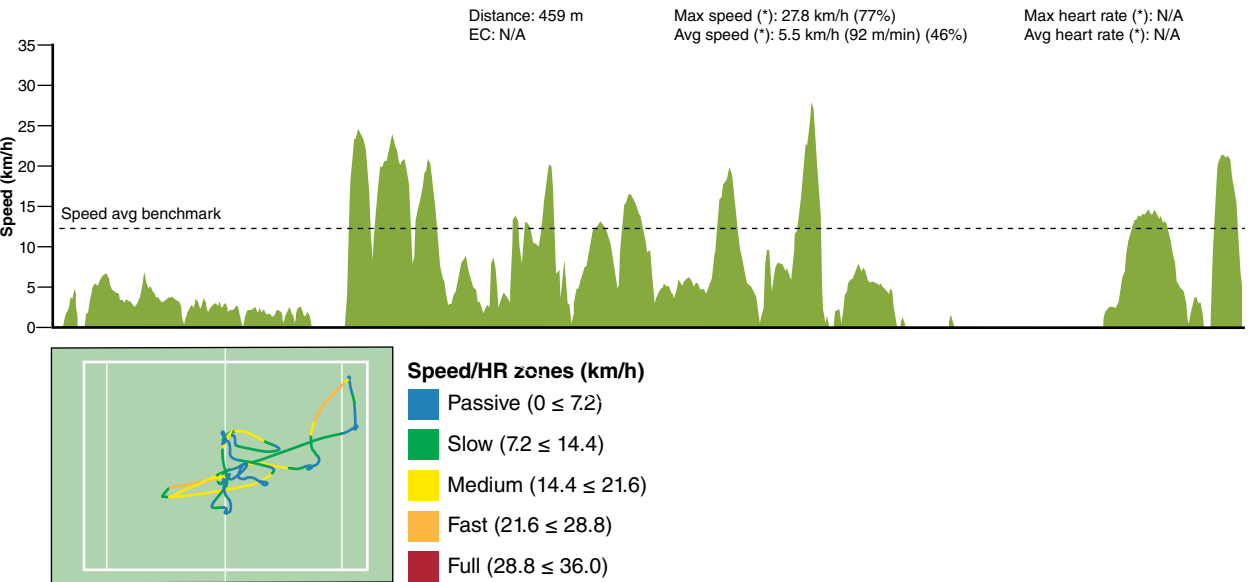


Figure 28.4 International Women's Rugby Match. The diagram illustrates what was identified as a very high 5 min period in a women's international rugby player's workload. In isolation, this speed profile tells us only part of the picture in determining overall workload. However, it does provide a valuable insight into what we can use to prescribe relevant bouts of high-speed running, repeated efforts, and their relationship to game-related skills. Finding the hardest and most intense periods in your sport can greatly assist the design of SSGs and other conditioning activities.

SSG 1: Modified Rugby League

Each SSG in its own right can be valuable when managing or replicating match intensity in training, along with the integration of core skills. For our modified Rugby League SSG (figure 28.5), we play 3 versus 3 across a 35 × 10 m area (from sideline to halfway across and halfway to 10 m line). The player must touch the sideline after making a tackle. Each side is allowed 6 tackles only, as per Rugby League rules. Players must rise to their feet after being tackled to tap the ball, and no kicking is allowed. The idea is to keep the games moving, and there are no stoppages. When a try is scored, the team is to immediately play on. The high neuromuscular loads experienced in this SSG, due to the changes of direc-

tion, short work durations, and short recovery periods, align with a type 4 HIIT target.

Correlation to Match Metrics

- Distance = low
- HI distance = low
- HI duration = low
- Sprints = moderate
- Impacts = high

Here we identified a high relationship between SSG impacts and game demands with a moderate relationship to in-game sprint metrics but a low relationship to high-intensity distance and durations.

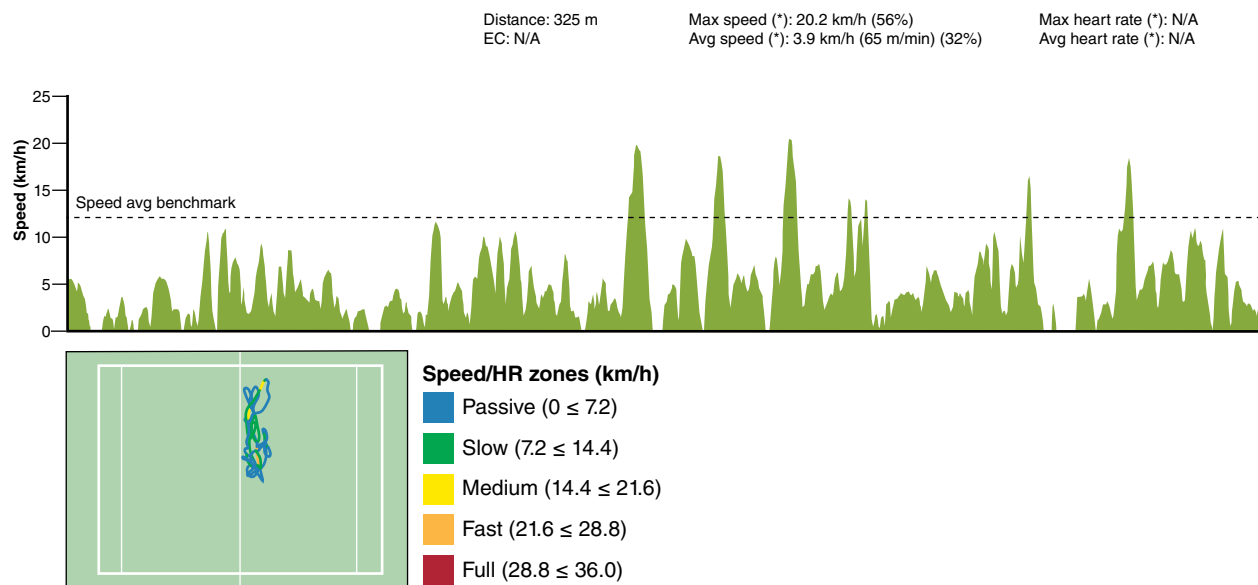


Figure 28.5 Modified 3 versus 3 Rugby League on a 35 × 10 m playing field.

chapters 29 and 30), so subtle manipulations in GBHIIT factors (see chapter 5) can help you achieve the stimulus you desire.

It is important to understand the game demands for the positional groups you are working with (figure 28.1). Figure 28.4 illustrates a GPS-derived game demand example and how simple GBHIIT sessions (figures 28.5 to 28.7) can be manipulated to achieve the appropriate training response at, below, or above the match demands as required. In these

examples, the metrics to determine game demands included:

- Distance covered (m)
- High-intensity (HI) distance covered (m over 15 km/h)
- Duration at high intensity (HI) (min:s over 15 km/h)
- Sprints (quantity over 20 km/h)
- High-intensity impacts (#)

Ssg 2: Follow Me, Modified Rugby Union

With this SSG, we identified its large relationship with match demands for high-intensity distance, duration, repeated high-intensity efforts, and sprint metrics. Follow me (figure 28.6) is played 12 versus 12 across the field on a 70×28 m area from sideline to sideline and halfway to the 22 m line. The rules are like Rugby Union, but you can tackle only your allocated opposite number (figure 28.1), although blocking is allowed. No kicking, tackles are unlimited, and rucks are uncontested (i.e., when the ballcarrier releases the ball, the opposition cannot touch it). When tackled, players pop off the ground, with defenders moving 5 m back. No stoppages occur to keep the game moving, and when a try is scored, play immediately moves on. Coaches encourage all players to work back

behind the ball. This game is closest to match demands when considering the number of repeated high-intensity efforts (RHIEs), and by adding an interchange rule it can be narrowed further. The game’s high neuromuscular and metabolic loads associated with its changes of direction, extended work durations, and short recovery periods allow this SSG weapon to hit the type 4 HIIT target.

Correlation to Match Metrics

- Distance = high
- HI distance = high
- HI duration = high
- Sprints = high
- Impacts = high

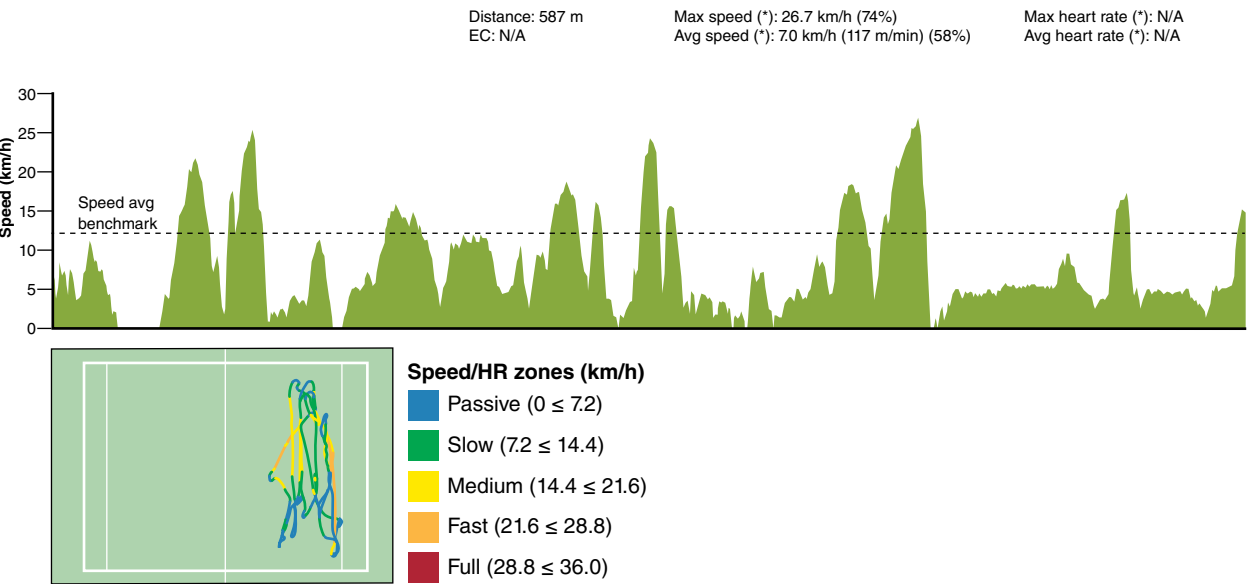


Figure 28.6 Modified 12 versus 12 Rugby Union on a 70×28 m playing field.

HIIT With Short Intervals

Throughout the world of high-performance sport, there seems to be good awareness of the importance of achieving certain volumes of high-speed running. Short intervals that include changes in direction (type 2 and 4 targets) are a good way to manage this need for high-intensity work while still managing the training stress. We prefer to keep the duration of

our efforts short, using short-interval weapons of 10 s to 20 s efforts on a 1:1 to 1:4 work:rest ratio (see figure 28.8).

Such intervals are used regularly during the season to provide extra work to individuals who need to maintain their levels of aerobic oxidative and anaerobic glycolytic power. In others, short intervals are used as top-ups at the end of training sessions

SSG 3: Down and Up, Modified Touch Rugby

This SSG we call down and up (figure 28.7) is played 12 versus 12 across the field on a 70×100 m area, from sideline to sideline and full field length. The game follows the basic rules of touch rugby, but as the name implies, all players must go to ground before returning to the opposition try-line after making a touch or being touched. All touches must be two-handed, and only six total are allowed. Kicks are allowed to hand only, with no bounce to regather. No stop-pages are used to keep the game moving, and when a try is scored, play immediately moves on. For this SSG, we identified a moderate relationship with match demands for high-intensity distance, duration, and sprint metrics, but a low

relationship to impacts. Again, there is high neuromuscular loading associated with changes of direction, work durations are extended, with short recovery periods aligning this SSG as a type 4 HIIT weapon. Importantly, impacts and the associated neuromuscular burden are low, so it can be used to facilitate recovery.

Correlation to Match Metrics

- Distance = high
- HI distance = moderate
- HI duration = moderate
- Sprints = moderate
- Impacts = low

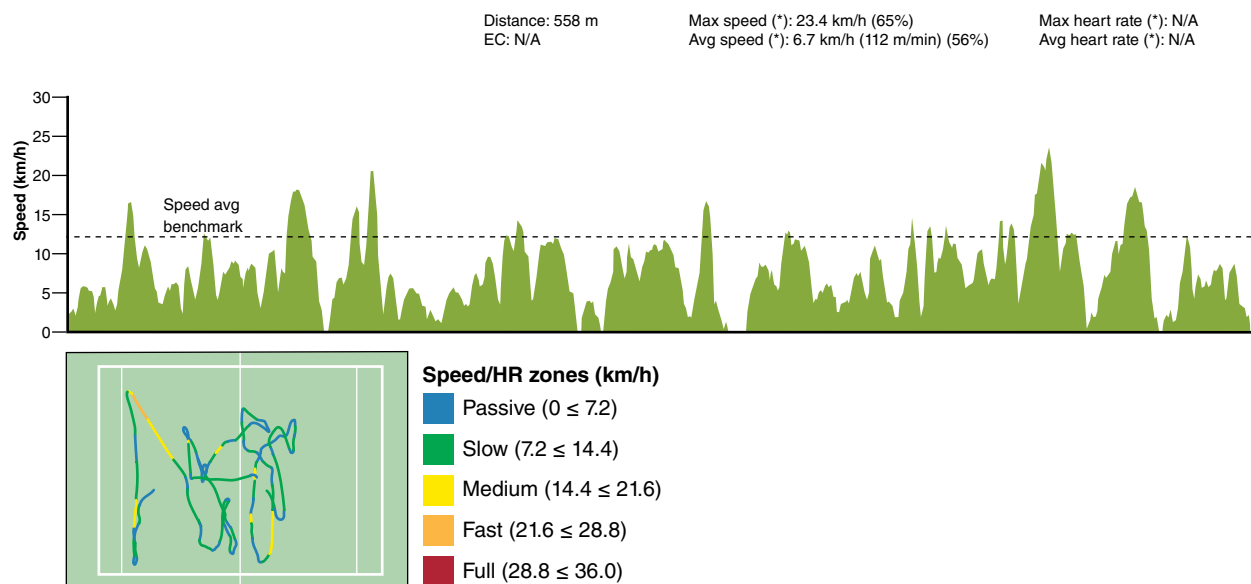


Figure 28.7 Modified 12 versus 12 touch rugby on a 70×100 m playing field (full size).

to provide additional volume to players who may be lacking fitness, have been exposed to minimal game time, or (due to injury, illness, or selection pressures) are training fully with the team but just not playing.

Typically we use a number of methods to determine the health and fitness of our squad. One such method is benchmarking, or monitoring, that occurs in-season, such as with timed shuttles or intervals, where we have acceptable position-related targets. For example, we might use a 150 m shuttle in 25 s

for outside backs with one turn (100 m then 50 m). Likewise, an intermittent bouncing to feet 88 m for tight forwards is used: the player starts on the ground, bounces to his feet, sprints 22 m, hits the ground, bounces to his feet again, and repeats for 88 m (4×22 m), which must be performed in less than 20 s.

The issue with doing large amounts of straight-line running and shuttles is of course the lack of specificity related to game performance and the resulting weak transference of such training, e.g.,

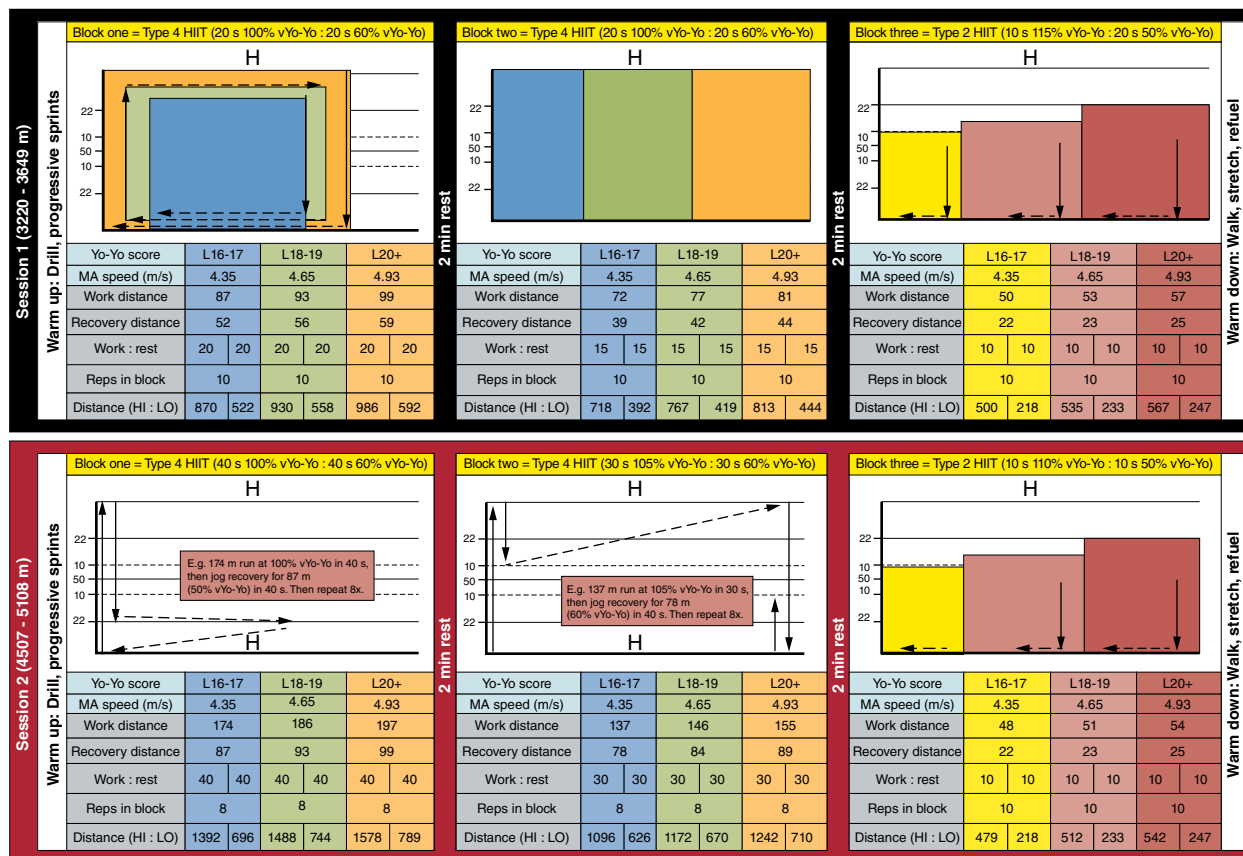


Figure 28.8 Examples of typical short-interval training weapons for elite Rugby Union preparation. Note the individualized approach based on vYo-Yo. Here specific conditioning targets (type 2, 3, etc.) are addressed by utilizing the range of interval training weapons shown.

the best shuttle runner does not necessarily translate to being the best player. To continue to improve relevance and transference, the inclusion of position-specific tasks, skills, and movements is often applied. For example, we may ask a player to complete a shuttle before having him work on some rugby-specific ball skill, or complete a taxing static task before, during, or after the running effort, e.g., ball wrestle, four consecutive tackles, and so forth.

While the optimal combination of high-speed running, deceleration, acceleration, static tasks, skills, metabolic, and locomotor stress is difficult to quantify, the guiding principle we use is to obtain certain target percentage levels of game stress for varied durations and volumes, e.g., 10 min of work for top-up or 45 to 60 min of work if a player is returning to play from a lengthy time on the sidelines.

Repeated Sprint Training

The use of repeated sprint training (RST) or repeated speed is highly relevant for backs and loose forwards.

The need to run fast during the game occurs often for these players and therefore developing speed, and repeated speed, with only short to moderate recovery is important. Typically this should be kept as game specific as possible, across distances of between 5 and 70 m, with rest periods of 3 to 4 times the duration of the effort. For these weapons, aiming for type 5 targets, the number of repetitions depends on the duration of the effort, such that if we are focused on training or improving speed off the mark (e.g., maximal acceleration) in a manner similar to a defensive line coming forward quickly (e.g., line speed), then we may have blocks of 6 to 10 efforts of 10 m with 15 to 20 s rest periods. Completing 3 to 6 blocks of this exercise can occur depending on what else needs to be included in the day's training. For example, this acceleration focus may then lead on to an attack focus in which we are now interested in working on 30 to 40 m efforts, the wing or full-back has a ball in hand, and is running fast but at the same time trying to beat a defender (1 versus 1).

This exercise might involve 5 to 8 repetitions on attack and defense to ensure adequate rest between attacking efforts.

In addition to RST on-feet, we also use metabolic training with similar work:rest intervals and intensities, but in off-feet type situations, e.g., using a rowing or bike ergometer. In this way we can add cardiovascular and metabolic stress (i.e., type 3 targets) but not necessarily add neuromuscular and locomotor stress associated with types 2, 4, and 5 targets, thus helping reduce the likelihood of injury that can occur with excessive high-speed running volume.

HIIT With Long Intervals

We believe that long-interval HIIT is important for all Rugby Union playing positions during the pre-season, as well as sometimes during the competition season for certain players (particularly for backs and loose forwards). Typically, these efforts are made up of 300 to 1200 m shuttles. Due again to the importance and regularity of deceleration and acceleration, these long intervals have a lot of turns involved, forming type 4 HIIT targets for us. During some parts of the season, these turns may require the player to go to ground and/or bounce to feet before continuing with the effort. Such modifications are made based on the individual's needs or requirements and current game load. These intervals will usually be on a varied work:rest ratio of 2:1 to 1:1.

Maximal aerobic speed (MAS) blocks are often prescribed in Rugby Union using many different formats (e.g., shuttles, field loops, etc.). Essentially, we prescribe targets using percentages of the vYo-Yo (calculated from YoYo test results; see figure 28.8). This method allows for target speeds to be prescribed around MAS, ensuring the appropriate stimulus is prescribed relative to the desired outcome.

Another common long-interval shuttle is the Bronco (1.2 km shuttle; 20 m, 40 m, 60 m out-and-back shuttle completed 5 times without rest), which is an effective and practical benchmarking run with applications in rehabilitation and return to play, hitting type 4 targets.

Depending on the athlete and the session priority, this type of long-interval session could be prescribed first (while fresh and to maximize running speed and performance) with other sessions (e.g., strength and power) scheduled after. However, if strength and power are the priorities for the individual, the priority session will occur in the morning and the session of secondary importance performed later in the day

(for further expansion on this issue, refer to chapter 6). The contents of each session, the purpose of each session, and the focus for the individual need constant consideration. Sequencing is vital to ensure desired adaptations occur and injury risk is reduced.

Sprint Interval Training

As Rugby Union players, it is important to be able to run fast! Line breaks and speed are one physical attribute that has been correlated with scoring tries; so it does appear that speed is king. To develop speed and sprint ability, it is common for us to use SIT to hit type 5 targets, with all-out sprints over 20 to 150 m (up to 30 s) at maximal effort and velocity, with long rests to ensure sufficient recovery and quality repeats (3 to 4 min; 1:8 work:rest). Depending on the positional group, the tight forwards are likely to not complete efforts longer than 20 m, but as we work through the loose forwards, inside backs, and outside backs, the distances (and ranges) increase.

Load Surveillance and Monitoring Tools

As already discussed, it is common practice to use GPS to track training load in elite Rugby Union. We find this to be a sound method of tracking locomotor load (e.g., distance, intensity, acceleration, deceleration, collisions, etc.) and predicting metabolic load. Many systems also allow for the simultaneous collection of heart rate data, and many elite teams use this information to inform their conditioning practices. While other aspects of load need consideration (see chapter 8), GPS data does offer insight into what is actually happening on the field and is our main marker of external training load. RPE is also an important consideration we use to assess the internal training load of the program.

There is a range of prescription ratios, percentages, and performance targets that are set within a team environment, but, while we believe that there is a place for the data, there is equally a requirement for common sense. The blend of the two is crucial. For example, hitting stable acute-to-chronic loading ratios is one thing, but not keeping an eye on how your players move so as to pick up obvious signs of movement competency, or lack thereof, is pure neglect. There is obviously the physical load of playing and training that needs consideration, but there are equally other stressors involved of importance, such

as travel (and associated sleep deprivation), individual game load, training age, positional stressors, and home life, on top of the psychological load experienced by players, including monotony, team meetings, media scrutiny, and, of course, the game.

Balancing stress (e.g., metabolic, psychological, performance, selection, etc.; see chapter 7) with recovery and relaxation strategies should always be prioritized to ensure optimal performance. Looking at one or two variables in isolation or becoming fixated on data over player well-being can be a dangerous practice. Time and energy must be directed toward practices that make a meaningful difference to athlete wellness that will translate into sharpness, as well as training and performance readiness.

Training Status Surveillance and Monitoring Tools

The training status monitoring tools that we've used in Rugby Union are generally limited to noninvasive measures and include wellness questionnaires and GPS/accelerometer-derived data, HR, and RPE responses to sessions. From a physical perspective (figure 28.2), it is common practice that players will undergo monitoring of strength, power, speed, and acceleration ability during certain times of the season. If an athlete is not strong enough relative to his history, ability, age, or position, then a strength focus may be implemented for the subsequent cycle. Similarly, if a player is carrying too much body fat, then extra work may be prescribed and nutrition habits addressed (see chapter 7). Similarly, if a player is not hitting the speed and acceleration numbers desired or expected, then there may be a shift toward more explosive movement focus in the program. From a fitness perspective, and specifically related to HIIT, the 1.2 km shuttle run (the Bronco mentioned previously) is often used to assess a player's fitness level. If fitness is not where it needs to be, then various HIIT weapons are prescribed accordingly to address identified deficiencies, depending on the time of the year, training cycle, as well as individual and positional needs.

Strategies for Structuring the Training Program Using HIIT

The inclusion of HIIT weapons into a rugby program can occur in many different ways for many different reasons. Within a season, there are numerous variables that we must balance to achieve consistent

high-level performance. Considering all of the important variables, not just a couple, ensures greater likelihood of success. In Rugby Union, we believe that traditional periodization approaches across all aspects of physical preparation can be unrealistic to achieve. The complexities of getting a 120 kg athlete lean, strong, powerful, light-footed, and fit with good ball skills and game understanding means that plans and periodization must be agile and adaptable. This flexible prescription implies considering the individual and team with respect to how workloads build toward peak targeted performance. The following section addresses some of the subtleties of a Rugby Union training program, specifically in relation to HIIT weapons.

Controllable and Uncontrollable Factors

Well in advance of most rugby seasons, we have a clear idea about who our opposition will be, when we will play them, where, and at what time. This schedule is locked in concrete and allows us to put our structures in place. However, once we have commenced our season, the balance between gym time, field time, meeting time, recovery time, travel time, home life, and relaxation becomes a moving beast and hence agile and adaptable programming and periodization becomes a necessity. In addition to this, the performance of the individual and the team requires the flexibility to adjust technical and tactical input if, and when, required.

Typically, the best way to ensure we are getting the metabolic, neuromuscular, and mechanical load we require for the team is to allow each individual to adapt and adjust often. If a game for a given player was very physical, with a relatively high number of tackles but relatively low high-intensity running distance meters, then our approach for the following week will immediately adapt from a preplanned program to accommodate the game exposure. Likewise, if we have traveled for 24 hours, the physical load of the subsequent week may change slightly to accommodate for individual differences in responses to the travel stress. Either way, it is important to ensure that each athlete is achieving a sufficient stimulus each week to maintain or improve physical capabilities.

Program Structure and Progression

In Rugby Union, the off-season and preseason typically follow a structured and periodized program mixing physical, technical, and tactical training

focuses, as there is no game to taper for nor is there a game to recover from. These phases are very easy to manipulate to target the desired mix of HIIT weapons. Bearing in mind we are preparing for upcoming matches, the need to introduce collisions becomes an important consideration during this phase.

During competition, the training week structure often becomes a reflection of how the team is performing and relates to player energy levels. Usually the training week becomes a balance between the need to recover from the game and to taper for the next game, and this is dramatically impacted by the turnaround time. Sometimes there may be as few as 3 or 4 d between matches and very little ability to complete physical work, so the emphasis must be on technical and tactical aspects. This constraint is not an issue for players involved in the games, but over time, can become an issue for those players not getting sufficient game exposure to enable them to be physically prepared as, and when, required.

To program and select the most appropriate HIIT target type for each player, we first must understand the landscape, that is, what has the player been exposed to from an overall stress perspective (physical, mental, etc.) and what is the next week likely to look like for him or her? It is important to understand the load experienced in skills, drills, and

training each day, and each week, to then be able to implement the right weapon, in the right amount, and at the right time! The physiological load of each individual session/day is generally easy to define, as we have planned in advance and we understand what it is we need. If there is a need to get more technical work completed on a specific day, then the ability to include somewhere in the day the appropriate weapon for each person is important. Some players who have high game exposure may simply have a recovery focus, while those players light on game exposure may be required to complete a variety of HIIT weapons on any given day.

Incorporation of HIIT

There are numerous ways to incorporate HIIT sequences into preparation weeks for Rugby Union, whether they occur within a technical or tactical session, at the end of a field-based session, or as a stand-alone session. Similarly, off-feet HIIT sessions before, during, or after strength and power work is common practice. The important consideration is whether individuals or the team require the stimulus and selection of the right weapon to address each player's moving target. This consideration ensures that the delivery of specific HIIT weapons is based on who needs what and when.

CASE STUDY

Nic Gill

While preparing a rugby team for a pinnacle game in the rugby calendar, we were required to travel 30 h 6 d before the away match. The match was to be played in front of 80,000 people supporting our opposition and we were to play in conditions 7 °C warmer than home and at altitude.

Six months earlier, we were informed of the challenge ahead, so the structures were put in place to allow us to do the impossible. In the 8 wk prior, while playing five other matches, we had planned to ensure our aerobic and anaerobic conditioning was reaching higher levels than previously. This required the prescription during bye weeks to include a range of HIIT weapons (see above) that ensured we had our bases covered. Rugby Union is about strength, power, speed, endurance, skill, and decision-making under pressure, so the need for the weapons to target long periods of play, as well as short intense periods of work, repetitively, with varied recovery times, was crucial to arm us with the arsenal needed to combat the enemy. We needed to ensure the physical stress of playing in the heat, and at altitude, did not inhibit our players' ability to perform their skills, execute their game plan, and make good decisions under pressure.

While the training load in the weeks between games was significant, we needed to ensure that we also dealt with the travel and environmental concerns. We did the work, gained confidence from the work, and dealt with what we were to experience. The need to recover from the matches, while improving our physical fitness over the 8 wk build, was a process of balancing on a knife's edge at times.

While the entire staff and players were put under enormous physical and technical pressure, we survived the battle to come out on top!